



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Innovative Teaching practice-NPTEL Video

Year/Semester/Section: III/V/A

Date:

Topic :Evaluation Measures

CO Coherency: CO4

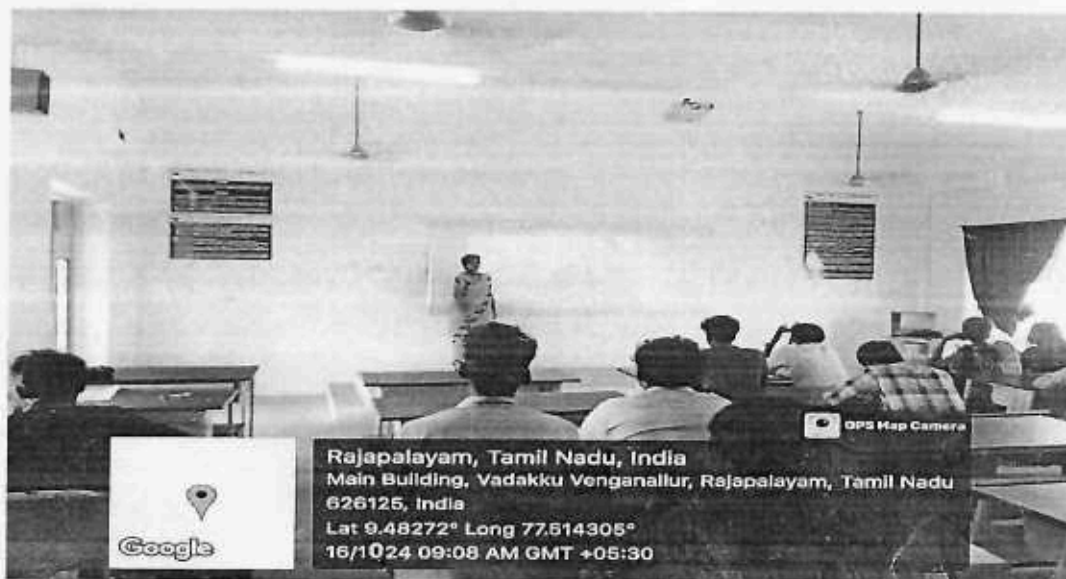
Link:

https://www.youtube.com/watch?v=sosZp0cUsIk&list=PL1xHD4vteKYVpaIiy295pg6_Sy5qznc77&index=45

Duration : 20:55 minutes

Video contains information about

- 1 to 4.02 – Miss classification error
- 4.03 to 9.33–Training parameters
- 9.34 to 10.48 –Active Learning
- 10.49 to 13.50 -Classifiers
- 13.51 to 17.30 -Variance parameter
- 17.31 to 17.46 –Bootstrapping



Theme:

The theme of evaluation measures in deep learning revolves around quantifying model performance in ways that align with the goals of the task at hand. These metrics serve as tools to assess how well a model generalizes, how accurate its predictions are, and how effective it is in solving a given problem. Evaluation measures allow practitioners to understand model strengths and weaknesses, and they provide a basis for comparing different models or algorithms.

Summary of Key Takeaways for Students:

- **Understanding Model Behavior:** Learn how different metrics help assess model performance in terms of error types, bias, generalization, and task-specific goals.
- **Practical Decision-Making:** Evaluate and select the right evaluation metric depending on the problem (e.g., classification, regression, generation, etc.) and specific goals (e.g., minimizing false positives, maximizing accuracy).
- **Improvement and Optimization:** Use evaluation measures to identify areas of improvement for deep learning models (e.g., tuning hyperparameters, addressing overfitting/underfitting).
- **Handling Real-World Challenges:** Learn how to handle challenges like **class imbalance**, **outliers**, and **model robustness**.
- **Clear Communication:** Understand how to present evaluation results effectively and interpret them in practical contexts (e.g., business decisions, healthcare applications, etc.).
- **Ethical Considerations:** Become aware of how evaluation metrics can impact model fairness, inclusivity, and overall ethical considerations in AI deployment.

In short, students learn to think critically about model performance and evaluation metrics, not just as numbers but as tools for guiding and improving machine learning projects in practical, ethical, and meaningful ways.

Feedback Questions

Did you find the additional content relevant to the topic?

☐ Yes

☐ No

Did the additional content enhance the understanding of the subject?

☐ Yes

☐ No

Would you like to see more content beyond syllabus in the future?

☐ Yes

☐ No

Did the extra content does not align with your learning goals?

☐ Yes

☐ No

Rate the topic



Pillay
Faculty Incharge

Wadhwa
HOD



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Department of Artificial Intelligence and Data Science

Academic Year 2024 – 2025 (Odd)

Degree, Semester & Branch: BTech/V/AD-A

Course Code & Title: AD3501-Deep Learning

Name of the Faculty member (s): R M Rajeshwari

Innovative Practice Description

- **Unit / Topic:** Unit IV- Evaluation Measures

- **Course Outcome:** CO4

- **Topic Learning Outcome:** TLO8

- **Activity Chosen:** Nptel Video-VIDEO LECTURE

- **Justification:**

- Understanding Model Behavior: Learn how different metrics help assess model performance in terms of error types, bias, generalization, and task-specific goals
- Learn how to handle challenges like class imbalance, outliers, and model robustness.

Time Allotted for the Activity: 45 Minutes

- **Details of the Implementation:**

Evaluation metrics are also essential for model improvement and optimization, guiding hyperparameter tuning and detecting issues like overfitting or underfitting. By analyzing low-performing areas, such as specific errors or patterns in the data, improvements can be made, like adjusting model complexity or gathering more data for underrepresented classes. Handling real-world challenges, such as class imbalance or outliers, requires a nuanced approach to metrics—precision, recall, and F1-score are often more informative than accuracy in imbalanced datasets, while metrics like MAE or Huber loss can handle outliers more effectively. Robustness, or the model's ability to maintain performance in the presence of noise or adversarial inputs, can also be evaluated with the right metrics. Models should be assessed for equitable performance across different demographic groups, ensuring that they do not discriminate or reinforce societal biases. Transparent reporting and an understanding of how metrics impact fairness and inclusivity are essential for ethical AI deployment. By carefully selecting, analyzing, and communicating evaluation metrics, we can make informed decisions, optimize model performance, and ensure ethical and fair AI systems.

- **CO – PO / PSO mapping:**

CO	PO 1	PO2	PO3	PO4	PO5
CO4	3	3	3	2	1

(1 – Low 2 – Moderate 3 – High)

- PO / PSO mapped:

Innovative practice	PSO1	PSO2	PSO3	-	-
	2	2	2	-	-
Justification for correlation	PSO1 (Model Evaluation) connects directly to understanding model behavior and practical decision-making for selecting the right evaluation metrics. It also ties into improvement and optimization, as evaluation results help identify areas needing refinement, such as overfitting or hyperparameter adjustments. PSO2 (Hyperparameters and Search) is connected to the improvement and optimization process, including handling challenges through fine-tuning models and searching for optimal parameters. Additionally, ethical considerations related to the fairness of the model also tie into hyperparameter tuning choices and adjustments. PSO3 (Data Performance Techniques) focuses on solving real-world challenges with proper data techniques and clear communication of results. This PSO also addresses how performance techniques influence ethical considerations, ensuring models are fair, unbiased, and applicable to diverse real-world scenarios.				

- Images / Screenshot of the practice: Report Attached
- Reflective Critique:
- Feedback of practice from students and other stakeholders: Separately Attached

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice) Report Attached

❖ **Challenges faced in implementation:**

- Implementing the **Evaluation Measure** is a **multifaceted challenge** that requires careful consideration of model evaluation metrics, real-world data challenges, and ethical implications. These challenges often require balancing technical, practical, and ethical considerations. Some potential solutions, such as automated tuning, fairness audits, and transparent communication, can help mitigate these challenges, but addressing them often requires domain expertise and iterative improvement of both the model and its evaluation strategies.

References:

- ❖ https://www.youtube.com/watch?v=sosZp0cUsIk&list=PL1xHD4vteKYVpaliy295pg6_SY5qznc77&index=45


Signature of Faculty Member

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Effective Date: 02.08.2021